

Internet use became the norm for humanity only very recently

Executive summary (what “recently” means, empirically)

A defensible way to test the claim that “internet use became the norm for humanity only very recently” is to operationalize “**norm**” as **majority adoption**—i.e., **more than 50% of the global population uses the internet** under a stated definition. On that framing, multiple long-run series converge on the same qualitative result: **global majority internet use is a post-2010s / early-2020s condition**, not a long-standing baseline.

- A widely cited long-run timeline reproduced by *Visual Capitalist* (sourced to **DataReportal**’s **Digital 2025 Global Overview Report** and **UN population data**) shows the global share online rising from **0.05% (1990)** to **40% (2015)**, then crossing majority by **2020 (57%)**, reaching **68% (2025)** (with absolute users rising from **2.6M (1990)** to **5.6B (2025)**) ([1]; [2]; [3]).
- In a more explicitly “official statistics” pipeline, the **ITU** headline estimate for **2025** is **74% online (~6.0B people)**, and the **World Bank WDI** world aggregate for `IT.NET.USER.ZS` also shows **74% in 2025**, reflecting ITU-sourced definitions and reporting channels ([4]; [5]).

A stricter interpretation—“**norm**” as **near-universal usage**—is **not yet true globally**. The ITU notes “universality” is conventionally treated as $\geq 95\%$ penetration; even its 2025 estimate (~74%) remains well below that threshold ([4]).

Bottom line: the internet became a **global-majority** phenomenon only recently, and it remains **far from universal**, with adoption still uneven across regions and incomes.

1) Definitions and measurement: what counts as an “internet user,” and why estimates differ

1.1 Two “norm” thresholds that answer different questions

- **Majority norm (this report’ s primary operational definition):** $> 50\%$ of the global population are internet users. This directly tests whether internet use is “typical” in a probabilistic, population-wide sense.
- **Universal norm (stronger standard):** $\geq 95\%$ penetration, the ITU’ s conventional “universality” benchmark ([4]).

These thresholds intentionally separate “most people” from “almost everyone,” which matters because global adoption has increased rapidly but remains uneven.

1.2 A comparability anchor: the ITU/World Bank/OWID “last three months” definition

For cross-country comparability, **Our World in Data (OWID)** uses the standard definition: an internet user is someone who accessed the internet **in the last three months**, from any location/device ([6]; [7]). The World Bank’s `IT.NET.USER.ZS` indicator uses ITU sourcing and metadata that explains historical shifts in measurement (operator data vs surveys) and timing mismatches that can affect year comparability ([8]).

1.3 Why DataReportal-style and ITU-style series can disagree (and how to handle it)

The report uses DataReportal-derived global aggregates because they provide widely referenced top-line totals and shares; however, **this evidence bundle does not include DataReportal/Kepios’ formal “internet user” definition details** (e.g., exact activity window, age thresholds, revision/backcasting policy). As a result, **the correct way to use these series is as complementary evidence**, while treating ITU/World Bank/OWID as the clearest definition-first benchmark ([2]; [6]).

2) Preconditions: why mass internet use was not “normal” in the early internet era

Internet connectivity began as specialized infrastructure—not as a mass household service—so it is analytically useful to separate (a) early technical feasibility from (b) public usability and (c) eventual mass adoption.

2.1 ARPANET and early institutional networking (small scale by design)

LiveScience’s timeline frames ARPANET as a research/military-funded system that validated **packet-switching** and enabled early applications such as **Telnet** (remote login). It also emphasizes the institutional scale: by **end of 1970**, ARPANET had **13 nodes**, and each site used specialized routing equipment (IMP) reported at **\$82,200** per unit ([9]).

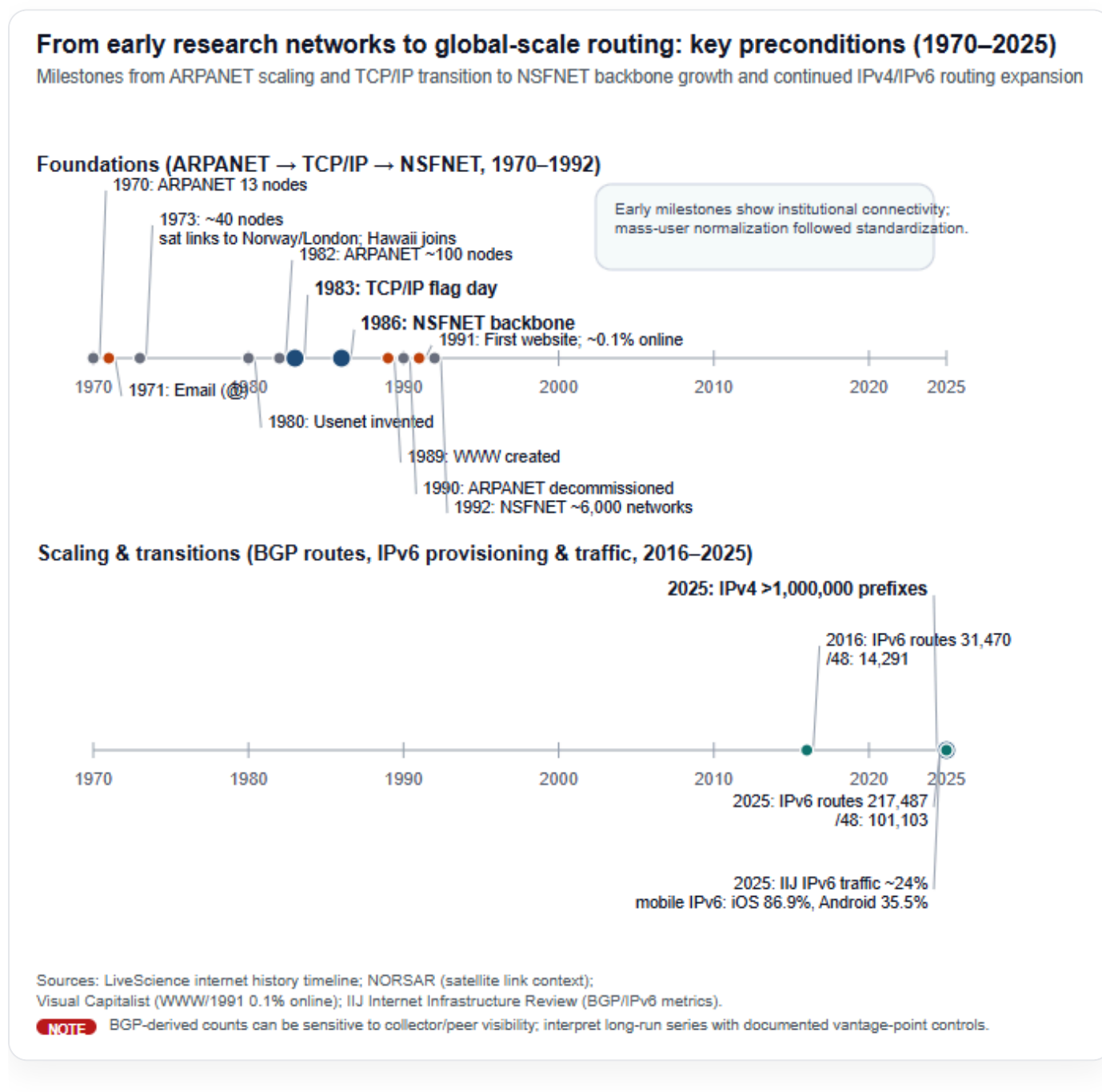
2.2 TCP/IP (1983) and NSFNET (1986–1992): scaling infrastructure before household normality

- The ARPANET switch to **TCP/IP on Jan 1, 1983** is widely described as the “birth of the modern internet” because it enabled the interoperable “network of networks” model ([9]).
- **NSFNET** (created **1986**) is described as a TCP/IP backbone linking supercomputing centers; LiveScience reports that by **1992**, **6,000 networks** were connected and **one-third** were overseas ([9]).

Interpretation: the core plumbing scaled globally before internet *use* became anything like a majority behavior—supporting the report’s central claim that “internet as the norm” is a much later social

outcome than the origin dates of internet protocols.

Figure (generated): technical and infrastructural preconditions timeline (ARPANET → TCP/IP → NSFNET → early web).



3) The World Wide Web (1989–1992): the usability breakthrough that made a public internet plausible

The internet predated the web, but the web's URL/HTTP/HTML stack—and portable clients—reduced the friction of finding and publishing information in a way that could plausibly scale beyond specialists.

3.1 A tight sequence of foundational web milestones

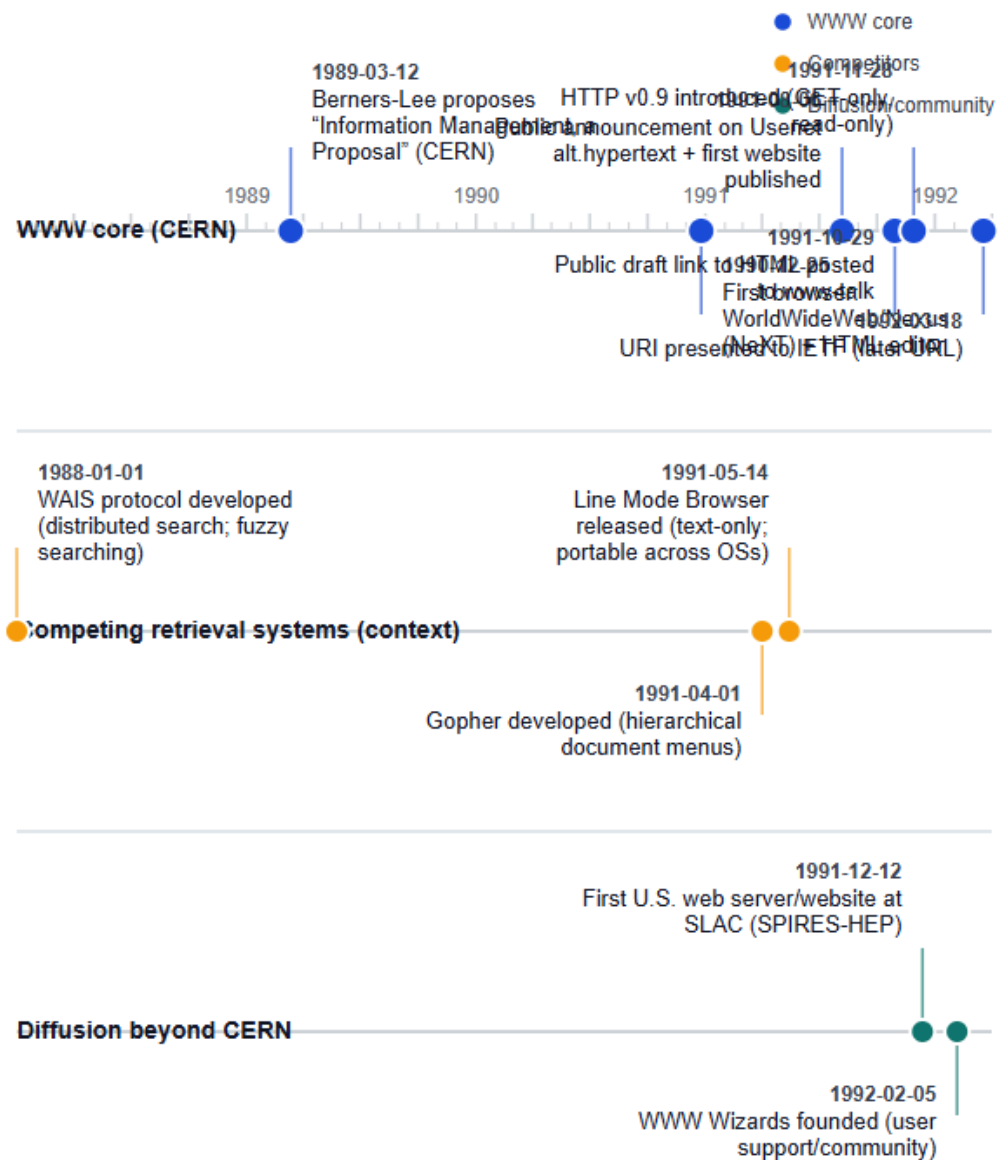
LiveScience documents a fast-moving sequence of events led by Tim Berners-Lee at CERN:

- **1989:** the initial proposal (“Information Management, a Proposal”), linked to early HTTP/HTML concepts ([9]).
- **Dec 25, 1990:** the first browser **WorldWideWeb/Nexus** for NeXT machines (with an HTML editor), with additional browser context summarized by *The History of the Web* ([9]; [10]).
- **Aug 6, 1991:** public announcement on Usenet `alt.hypertext` and publication of the first website ([9]).
- **Late 1991–1992:** early standardization milestones including public HTML drafts, **HTTP v0.9** (single `GET` method), and **URI→URL** presentation to the IETF ([9]).

Visual Capitalist connects this usability breakthrough to population reality: it notes the first website in **1991** when only **~0.1%** of the world was online, reinforcing how “public web availability” and “internet-as-normal” are separated by decades of diffusion ([1]).

The World Wide Web usability breakthrough (1988–1992)

A tight sequence of releases and standards turned the internet into a browsable, link-based public information system.



Interpretation: Within ~3 years, the web moved from proposal → browser → public release → early standards and diffusion nodes.
Sources: LiveScience Internet history timeline; TheHistoryOfTheWeb (browsers; Gopher; SLAC). Dates shown as reported in sources.

3.2 Why the web won mindshare: portability and competing information systems

LiveScience places the early web in a competitive ecosystem of information retrieval protocols:

- **WAIS (1988):** described as enabling distributed search (including fuzzy search) but "ultimately loses out to the web" ([9]).

- **Gopher (Apr 1, 1991):** a hierarchical menu-based alternative; *The History of the Web* provides a counterfactual framing of “what the web could have been” under Gopher-like constraints ([9]; [11]).
- **Line Mode Browser (May 14, 1991):** highlighted as easy to port across operating systems and “incredibly popular” despite limited features—evidence that cross-platform availability mattered early ([9]).

3.3 Early diffusion beyond CERN: institutional footholds and support communities

LiveScience describes early U.S. adoption via a first web server at **SLAC** (Dec 12, 1991), linked to the SPIRES-HEP database (“tens of thousands” of particle physics papers), and the creation of the **WWW Wizards** support group (Feb 5, 1992) ([9]; [12]).

Section takeaway: the web’ s key contribution to the “recent norm” story is not that it instantly made the internet mainstream, but that it created a **portable, public-facing layer** in the early 1990s—when global usage was still negligible.

4) The adoption curve: from near-zero to majority-of-humanity (1990–2026)

4.1 Long-run global milestones (DataReportal + UN population baselines, via Visual Capitalist)

The DataReportal/UN-benchmarked timeline reproduced by *Visual Capitalist* provides a compact, auditable set of waypoints (users; share of global population online):

- **1990:** 2.6M (**0.05%**)
- **1995:** 39.2M (**0.7%**)
- **2000:** 361M (**6%**)
- **2005:** 1.0B (**15%**)
- **2010:** 1.9B (**27%**)
- **2015:** 3.0B (**40%**)
- **2020:** 4.5B (**57%**)
- **2025:** 5.6B (**68%**)

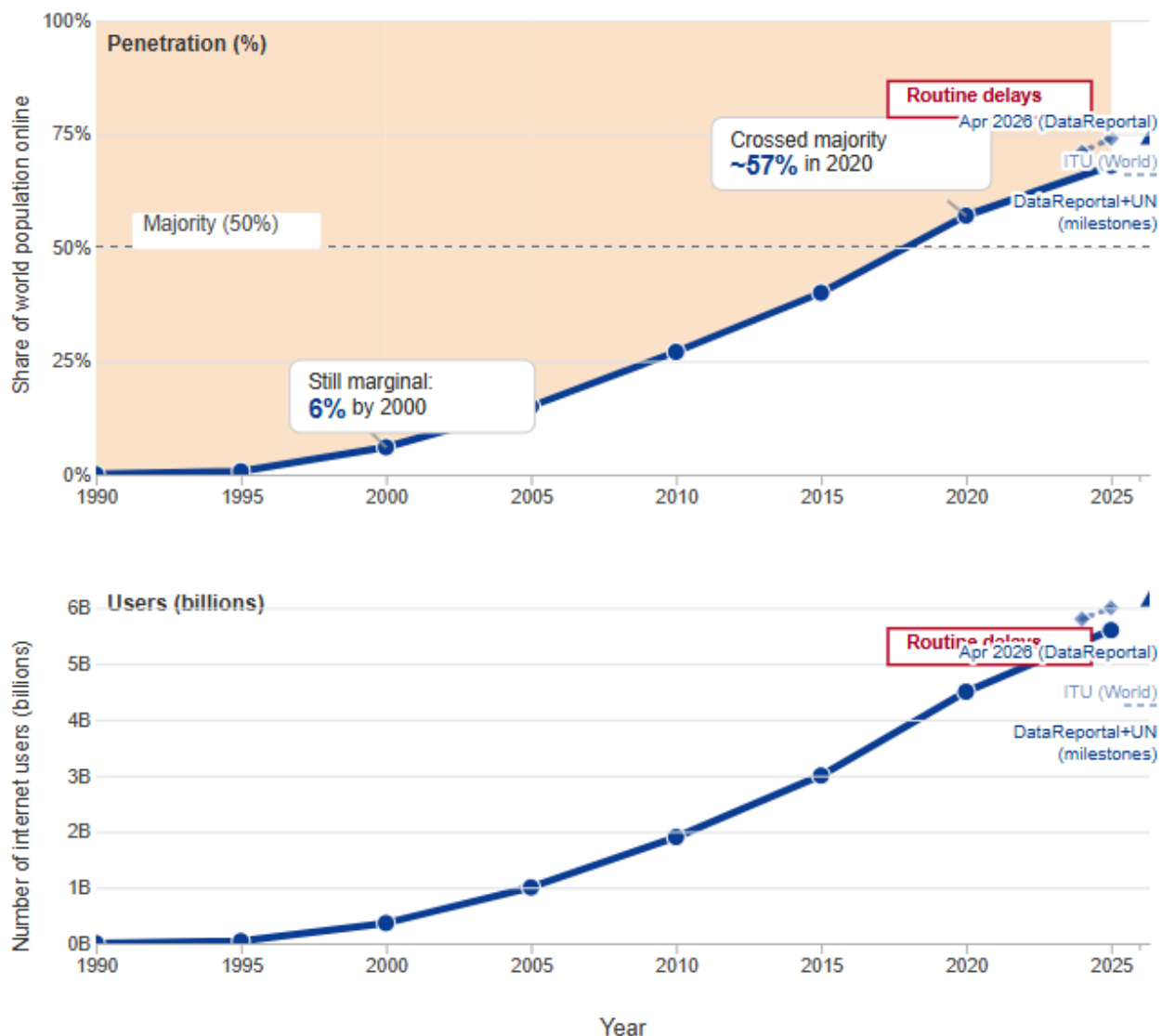
([1]; [2]; [3])

Implication for the topic: even after the web’ s invention, the internet was still far from “normal” globally: in **2000**, this series reports only **6%** online. Majority status appears only by **2020 (57%)**, supporting the claim that global normalization is very recent.

Figure (generated): global adoption curve with the majority (50%) threshold highlighted.

The adoption curve: how quickly “near-zero” became “most people” (1990–Apr 2026)

Internet user = accessed in the last 3 months (ITU/OWID-aligned definition); selected reported milestones



Sources: Visual Capitalist (DataReportal Digital 2025 + UN population) for 1990–2025 milestones; ITU Facts & Figures for 2024–2025. DataReportal Global Digital Overview for Apr 2026.

Note: Different compilers/updates can yield different 2025 levels; interpret direction and timing of majority adoption as robust.

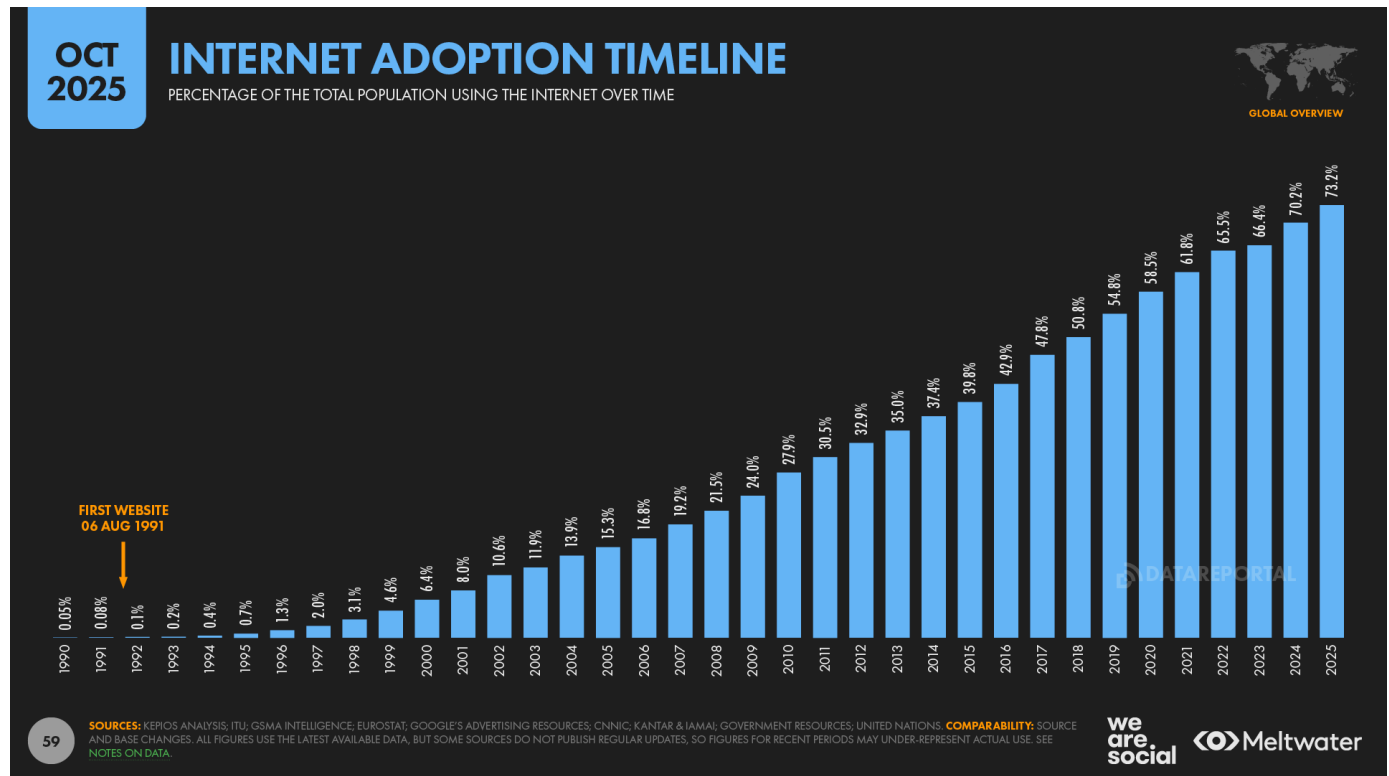
4.2 An ITU-aligned benchmark: “about three-quarters” online by 2025

The ITU reports **74% of the world online in 2025 (~6.0B people)** ([4]). The World Bank world aggregate for `IT.NET.USER.ZS` also shows **74% in 2025**, reflecting the same ITU-aligned definition pipeline ([5]). OWID provides a consistent narrative anchor: **“Only 63% of the world’s population was online in 2023”** ([6]).

How to read differences (without overclaiming): point estimates can differ across series (e.g., DataReportal-based vs ITU/WDI), but **both place “majority online” as a very recent milestone** and both situate the mid-2020s as a high-but-not-universal plateau.

4.3 April 2026 snapshot: internet use is “the norm” by majority share, but not complete in headcount

DataReportal’ s global digital overview reports **April 2026** at **6.12B internet users (73.8% of world population)**, leaving **2.17B unconnected**, and explicitly cautions that “**routine delays**” may understate recent growth ([13]).



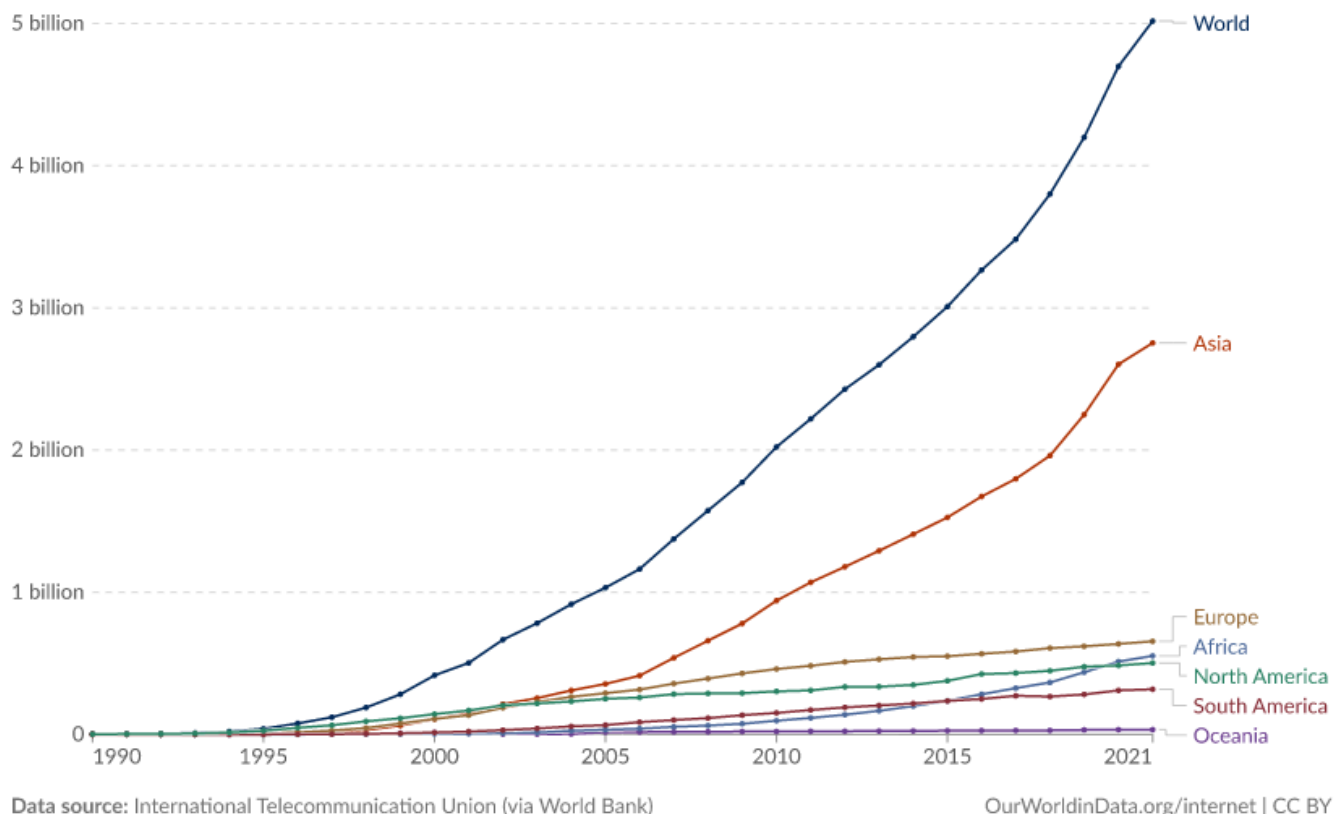
4.4 The “recentness” also shows up in absolute numbers (OWID visualization)

OWID’ s “**Number of people using the Internet**” chart (defined as people who used the internet in the last three months) visually emphasizes that **most absolute growth is concentrated in the last ~15 years**, consistent with the milestone-based interpretation above ([14]).

Number of people using the Internet

Our World
in Data

Number of people who used the Internet in the last three months.

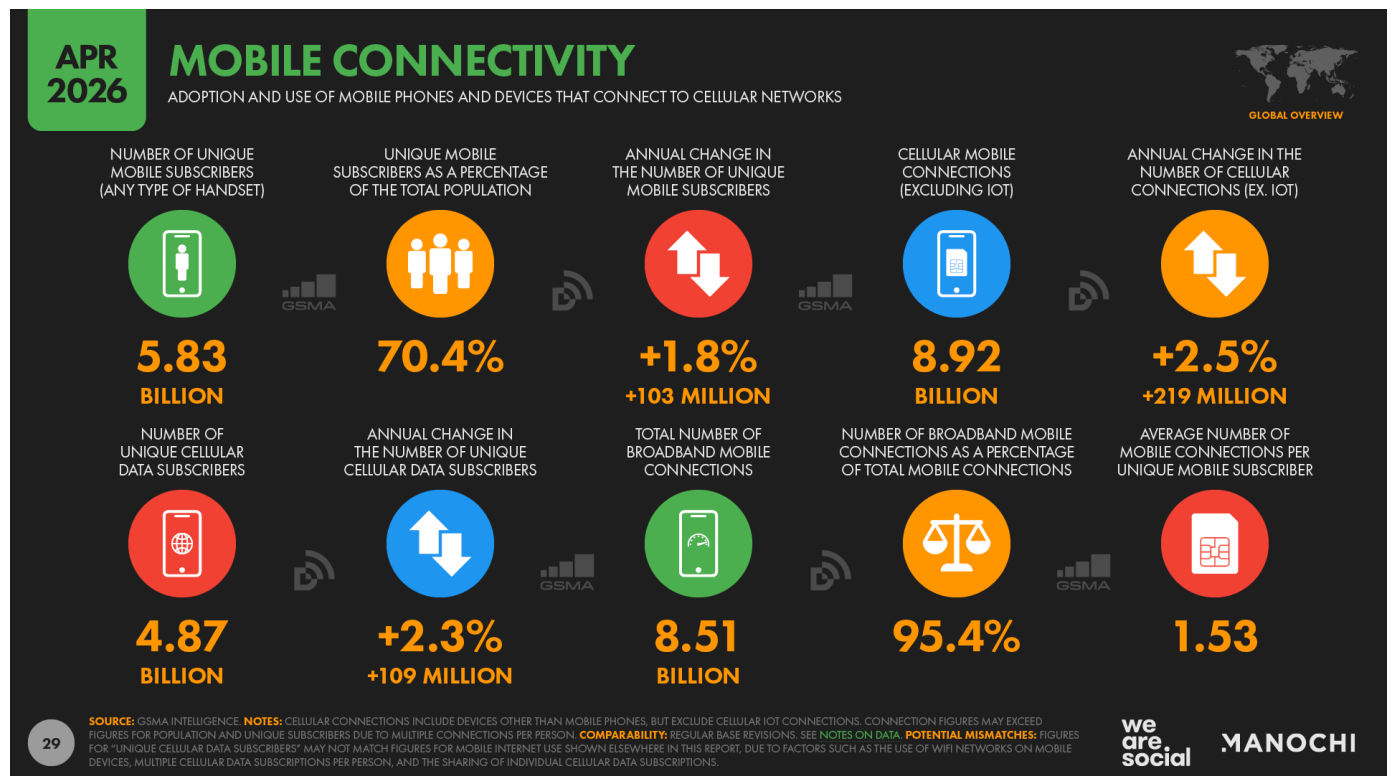


Section takeaway: whether viewed as share-of-population or absolute users, the curve's steepest climb happens late, and "internet use as a global majority behavior" is a development of the late 2010s/early 2020s, not the early web era.

5) Why adoption accelerated so late: mobile-first access and "usage-value" complements

5.1 The internet became normal primarily through phones (not fixed PCs)

Visual Capitalist reports that **96% of internet users accessed via mobile phones in 2025**, tying late acceleration to smartphone-centered access and falling barriers to connectivity ([1]). DataReportal's April 2026 snapshot similarly reports **96.2%** of internet users access via a mobile phone "at least some of the time" ([13]).



DataReportal also reports behavioral reinforcement: **mobile accounts for 51.6% of global web traffic (April 2026)**, suggesting mobile-first is not just access availability but a dominant usage channel ([13]).

5.2 Scale indicators (and denominator hygiene)

DataReportal distinguishes "people," "devices," and "identities," which is essential when interpreting mass adoption:

- **5.83B unique mobile users (70.4% of global population)**
 - **7.64B smartphones in use**
 - **smartphones are ~89% of mobile phones**
- ([13])

This is consistent with the report's core narrative but must be handled carefully: "unique mobile users" and "smartphones in use" are not the same metric as "internet users."

5.3 Online services become "must-have" once connectivity is widespread

DataReportal reports **5.79B active social user identities (69.9% of the world)** and that **94.7% of internet users use social platforms monthly** (with GWI estimates of **96.7%** for adults 16+ in larger economies) ([13]). These figures are best interpreted as evidence that, once connected, a large share of users engage with major internet-native services—potentially reinforcing the perceived value of being online—while respecting that "identities" are not guaranteed unique individuals.

Section takeaway: the timing of "internet as the norm" aligns with mobile-first diffusion and the maturation of widely used online services, rather than with early protocol inventions.

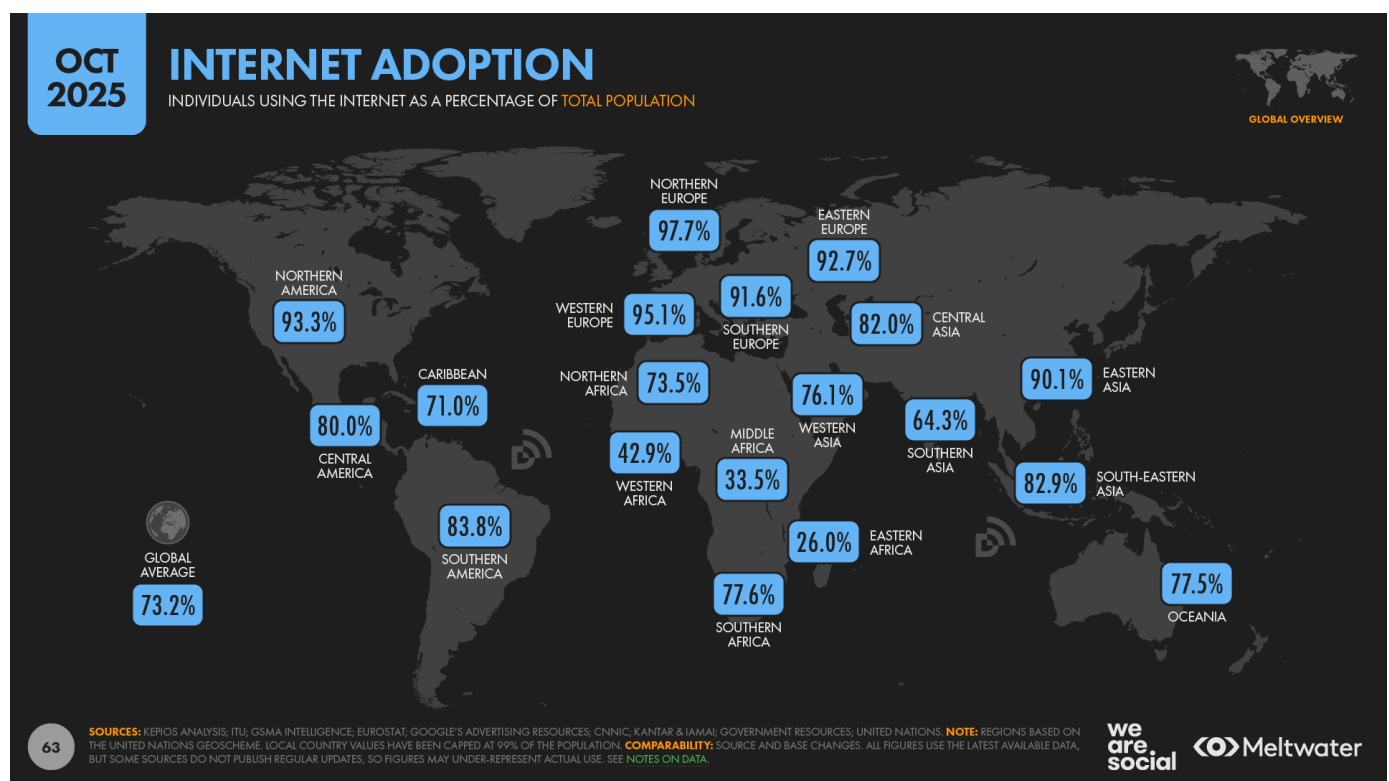
6) “Norm” is not universal: who remains offline, and where “majority” still does not hold

6.1 Billions remain offline even after a global majority is online

The *Visual Capitalist* synthesis reports **5.6B users (68%) in 2025** and **2.6B unconnected**, emphasizing that “majority” still implies a very large offline population ([1]). DataReportal’s April 2026 snapshot reports **2.17B unconnected** ([13]).

6.2 Offline status is geographically concentrated

DataReportal states unconnected populations are concentrated mainly in **Southern Asia and Africa** ([13]).



6.3 A diagnostic shift: “usage gap” can dominate “coverage gap”

GSMA’s *State of Mobile Internet Connectivity 2024* distinguishes:

- **Coverage gap:** 4% of the global population not covered by mobile broadband
 - **Usage gap:** 39% covered but not using mobile internet
- ([15])

Interpretation: where coverage is increasingly widespread, the binding constraints for the “next wave” can shift toward adoption frictions (affordability, devices, skills, relevance). This report does not attempt to quantitatively decompose those drivers by country because the necessary harmonized codebooks and integrated indicator panel are not fully present in the supplied evidence.

6.4 Regional and income divides (ITU-linked comparisons)

The *Visual Capitalist* summary (citing ITU comparisons) reports large regional gaps: **Africa 38%** vs **Americas 87%** and **Europe 91%**, and income gaps: **93% online in high-income countries** vs **27% in low-income countries** ([1]; [16]).

A time anchor for persistence appears in the ETTG annex (drawing on ITU facts and figures): **Africa 25%→38% (2019→2024)** vs **Europe 82%→91%**, while the world rose **53%→68%** ([17]).

6.5 Within-region divides: youth and urban/rural (Africa vs Europe, 2024)

The same ETTG/ITU-based annex reports that in 2024:

- **Youth vs rest:** Africa **53% vs 34%**; Europe **98% vs 90%**
 - **Urban vs rural:** Africa **57% vs 23%**; Europe **93% vs 86%**
- ([17])

6.6 Device capability constraints in Sub-Saharan Africa (GSMA excerpts)

GSMA excerpts in the annex also show a device mix consistent with capability barriers:

- **Feature phones (2019):** Sub-Saharan Africa **40%** vs Europe & Central Asia **11%**
 - **4G smartphones (2019):** Sub-Saharan Africa **22%** vs Europe & Central Asia **52%**
 - **4G smartphones (2023):** Sub-Saharan Africa **33%** vs Europe & Central Asia **55%**
- ([18])

Section takeaway: the world crossed a global-majority threshold only recently, but large regional, income, and within-country divides mean “internet use is normal” remains context-dependent—and is still not a majority experience in some places.

7) Methodology and verification guide (how to audit a “recent norm” claim)

This section consolidates the report’s triangulation logic into a reviewer-friendly checklist, with explicit limits where the evidence base is incomplete.

7.1 Minimum requirements to assert “internet use became the norm recently”

1. **State the definition of “internet user.”** For maximum comparability: ITU/World Bank/OWID (“used in the last three months”) ([6]; [8]).
2. **State the threshold.** This report uses **>50%** for “majority norm,” while documenting that ITU’s “universality” benchmark is **≥95%** ([4]).

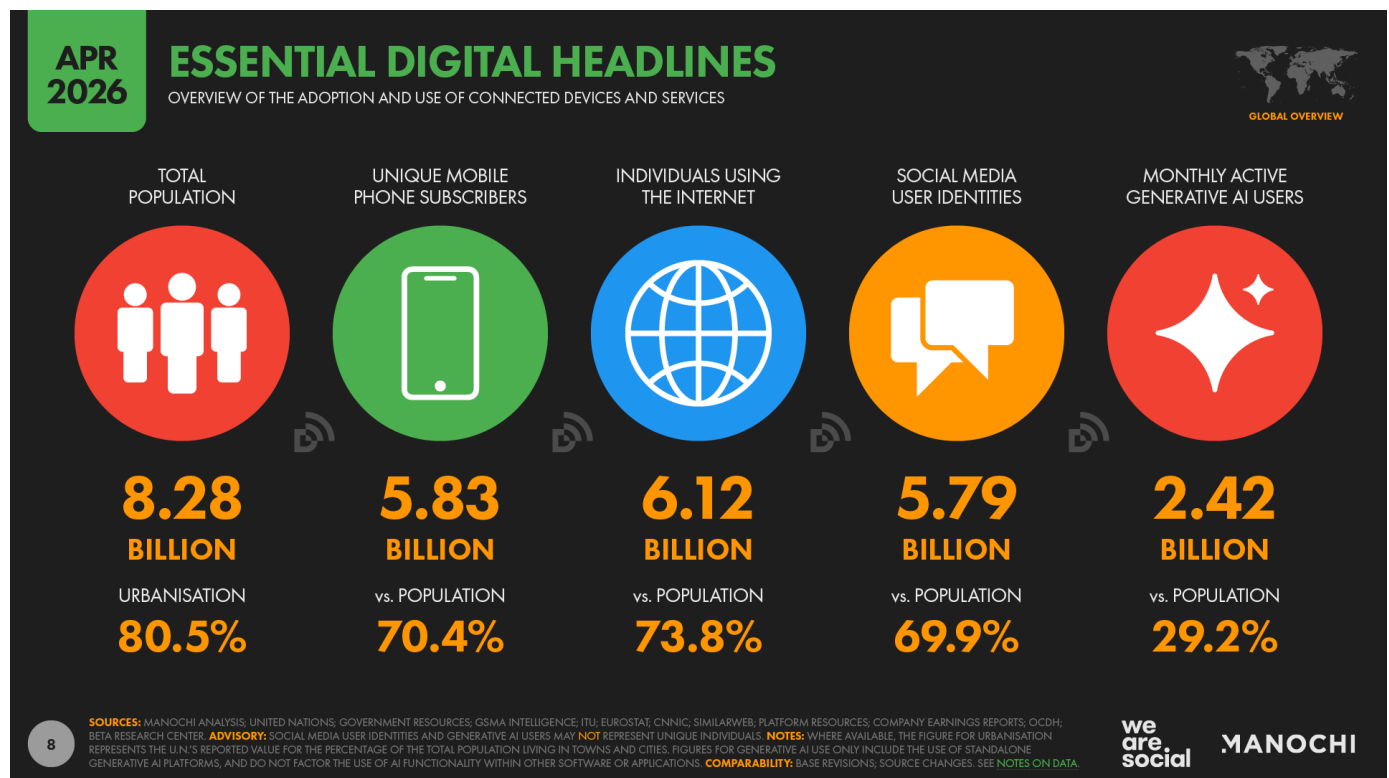
3. **Compute the crossing year from a year-by-year series.** Do not infer an exact crossing year from a handful of infographic points; use a downloadable dataset such as OWID's "share" series or World Bank WDI export ([7]; [19]).

7.2 Triangulation approach used in this report (and what it does *not* claim)

- **Primary definition-first anchor:** ITU-aligned `IT.NET.USER.ZS` / OWID "share" (last-three-months usage) ([6]; [19]).
- **Long-run narrative and milestone points:** DataReportal+UN benchmarks as reproduced by Visual Capitalist (useful for communicating the speed of diffusion, while keeping definition equivalence caveated) ([1]; [2]; [3]).
- **Late-stage adoption friction diagnostics:** GSMA "coverage gap vs usage gap" framing (mobile-internet specific) ([15]).

7.3 Reference-date and revision risk (especially for near-real-time dashboards)

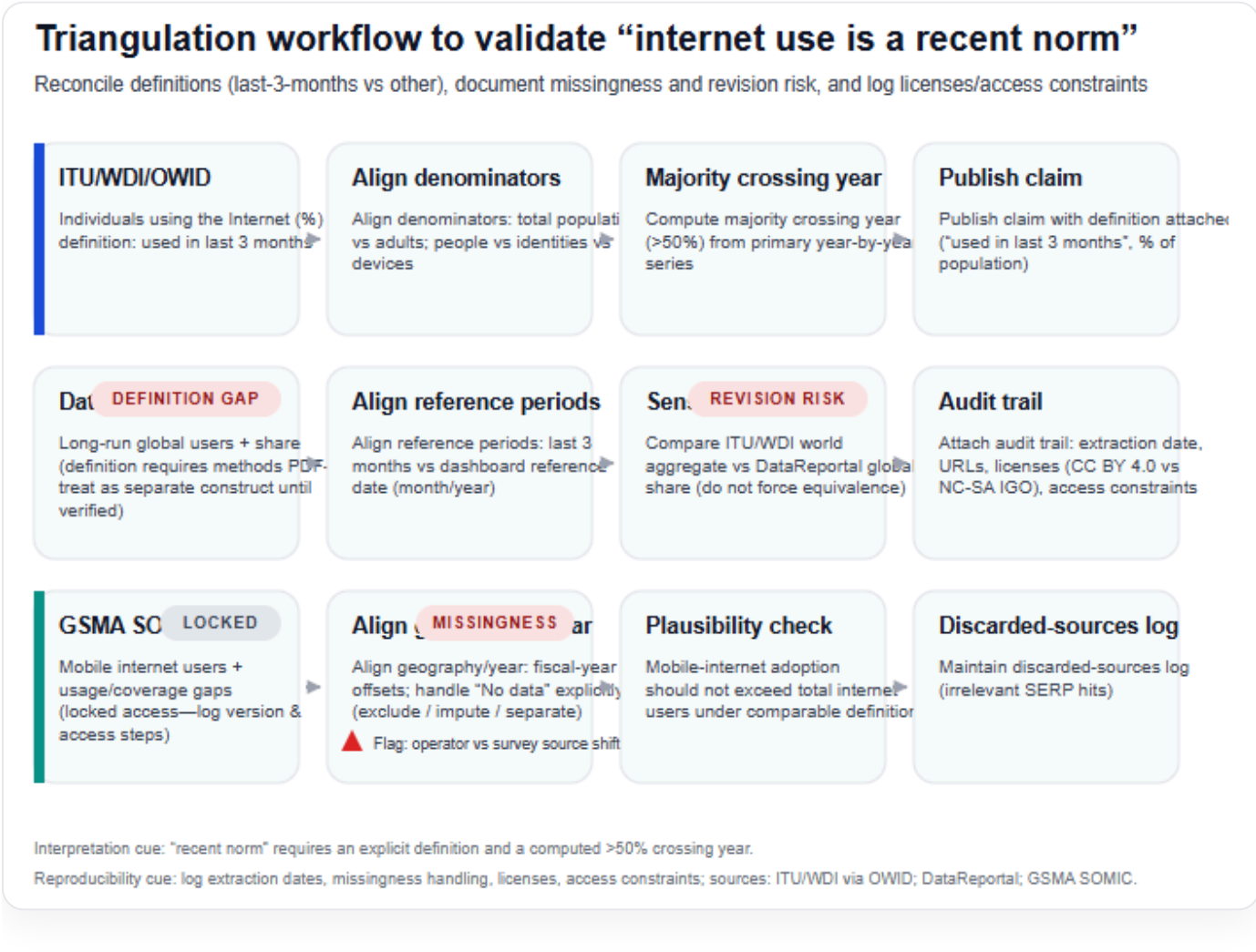
DataReportal's global overview explicitly warns of "routine delays," meaning recent totals can be revised; therefore, any claim should log the **reference month (e.g., April 2026)** and the page/version used ([13]).



7.4 Denominator hygiene: avoid category errors

DataReportal's reporting includes constructs such as "active social user identities," "unique mobile users," device counts, and internet users. Analyses must not treat these as interchangeable; they answer different questions ([13]).

Figure (generated): reviewer checklist for auditable “recent norm” statements (definition → threshold → dataset → timestamp).



Conclusion

Under a transparent “majority adoption” definition of “norm,” the evidence supports a precise and verifiable claim: **internet use became normal for humanity only very recently**, with long-run milestone series placing the world below **50%** as late as **2015 (40%)** and above **50%** by **2020 (57%)** in the DataReportal/UN-benchmarked timeline ([1]). ITU/World Bank/OWID series place the mid-2020s at roughly **three-quarters online** (e.g., **74% in 2025**)—high, but still far below “universal” usage ([4]; [5]; [6]).

The same evidence base also clarifies the boundary condition: a global majority can coexist with **billions still offline**, and adoption remains highly uneven across regions and incomes (e.g., Africa vs Europe and high- vs low-income comparisons via ITU-linked reporting) ([16]; [17]; [1]).

References (links used in-text)

- Visual Capitalist — “Visualized: The Growth of Global Internet Users (1990–2025)” (sourced to DataReportal + UN): [1]
- DataReportal — Digital 2025 Global Overview Report: [2]
- UN Data Portal — population baseline used in the Visual Capitalist/DataReportal timeline: [3]
- DataReportal — Global Digital Overview (Digital Around the World): [13]
- ITU — “Internet use” facts & figures (2025): [4]
- ITU — Internet-use comparisons (2024 page referenced for regional/income splits): [16]
- World Bank — `IT.NET.USER.ZS` indicator: [19]
- World Bank — WDI metadata for `IT.NET.USER.ZS` : [8]
- Our World in Data — Internet topic page: [6]
- OWID Grapher — Share of individuals using the internet: [7]
- OWID Grapher — Number of internet users: [14]
- LiveScience — Internet history timeline: [9]
- The History of the Web — early browsers: [10]
- The History of the Web — Gopher alternative path: [11]
- The History of the Web — SLAC first U.S. website context: [12]
- GSMA Intelligence — State of Mobile Internet Connectivity 2024: [15]
- GSMA — SOMIC materials portal (as referenced in the ETTG annex excerpt): [18]
- ETTG — “From Gaps to Goals: Digital Transformation in Africa” (Data Annex): [17]

References

1. [Visual Capitalist](#)
2. [DataReportal Digital 2025](#)
3. [UN population portal](#)
4. [ITU “Internet use” \(2025\)](#)
5. [World Bank `IT.NET.USER.ZS` \(World\)](#)
6. [OWID Internet topic page](#)
7. [OWID “Share of individuals using the internet”](#)
8. [WDI metadata](#)
9. [LiveScience timeline](#)
10. [The History of the Web — early browsers](#)
11. [The History of the Web — Gopher alternative](#)
12. [The History of the Web — SLAC](#)
13. [DataReportal global overview](#)
14. [OWID Grapher — number of internet users](#)
15. [GSMA SOMIC 2024](#)
16. [ITU 2024 internet-use comparison page](#)
17. [ETTg annex](#)
18. [GSMA SOMIC materials portal](#)
19. [World Bank indicator](#)